

Digital Operations

Field Telemetry Ingestion Platform

Solution Architecture Document

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SENSITIVITY CLASSIFICATION: Internal Use

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Revision History

Version	Date	Changed By	Description
0.4	2026-07-09	K. Nwosu	Draft for architecture review

1. Solution Overview

The Field Telemetry Ingestion Platform collects measurement data from approximately 310,000 field devices · smart meters, substation sensors and line monitors · and makes it available for operational analytics, outage prediction and regulatory reporting. Peak ingestion is roughly 45,000 messages per second during a network event.

The platform replaces a set of per-region collection scripts with a single ingestion pipeline. It is a greenfield build on the operations Kubernetes estate.

2. Architecture

2.1 Ingestion Path

Devices publish to a regional MQTT broker over the private cellular APN. A set of ingestion workers subscribes to the broker, validates each payload, normalises units and writes to the time-series store. The workers hold no state between messages and scale horizontally on queue depth; any worker can be terminated at any point without loss of in-flight work, as unacknowledged messages return to the broker.

2.2 Transport

Telemetry is transmitted over plain MQTT without transport encryption. The devices are battery-constrained and the TLS handshake measurably shortens field life; the cellular APN is a private network isolated from the public internet, which we consider sufficient protection for measurement data. Device authentication uses a per-device pre-shared key issued at commissioning.

2.3 Analytics Access

The analytics service requires customer account context to attribute consumption. It connects directly to the billing system's Oracle database using a dedicated read-only account and a nightly query against the account and premise tables. This avoids the latency of the integration platform and was agreed with the billing team as a pragmatic interim measure.

3. Data Design

3.1 Message Format

Payloads are JSON. The schema is documented on the team wiki. Fields are added as new device types are onboarded; consumers are expected to ignore fields they do not recognise. There is no schema registry and no version field in the payload · the team's view is that additive-only changes make

versioning unnecessary.

3.2 Classification and Retention

Telemetry is classified Internal. Raw messages are retained in the time-series store for 90 days, after which they are aggregated to hourly summaries and the raw records are deleted by an automated purge job. Aggregates are retained for seven years to meet the regulatory reporting obligation. Deletion is logged and the job's output is retained for audit.

3.3 Physical Model

Object Type	Convention	Example
Table	snake_case, plural noun	device_readings
Column	snake_case, singular noun	reading_timestamp
Index	ix_<table>_<columns>	ix_device_readings_device_id

4. Failure Handling

Messages that fail schema validation are written to the application log with the offending payload and discarded, so that a malformed device firmware release cannot back up the pipeline. Transient write failures against the time-series store are retried twice; if both retries fail the message is dropped and a counter is incremented.

Duplicate delivery is possible because the broker guarantees at-least-once. The ingestion worker writes using the device identifier and reading timestamp as a compound key, so a repeated message overwrites rather than duplicates.

5. Deployment and Operations

5.1 Packaging

Workers are packaged as container images and deployed to the operations Kubernetes cluster via the standard pipeline. Configuration is supplied through environment variables and Kubernetes secrets.

5.2 Monitoring

Workers write plain-text log lines to standard output, which the platform's default collector forwards to the central log store. Queue depth and message throughput are exported as metrics and alert on threshold.

5.3 Availability

The platform runs across two availability zones. Brief ingestion outages are tolerable because devices buffer locally for up to four hours and replay on reconnection.

6. Open Items

- Agree the long-term approach for account attribution with the billing team
- Assess whether device firmware can support TLS within the battery budget
- Confirm the operational owner for the ingestion pipeline